1) Test-Connection and nslookup

```
PS C:\Users\guruv> Test-Connection www.google.com
>>

Source      Destination     IPV4Address     IPV6Address                    Bytes    Time(ms)
------      -----------     -----------     -----------                    -----    --------
DESKTOPGV   www.google.com  142.251.221.100                                 32       23
DESKTOPGV   www.google.com  142.251.221.100                                 32       43
DESKTOPGV   www.google.com  142.251.221.100                                 32       104
DESKTOPGV   www.google.com  142.251.221.100                                 32       24


PS C:\Users\guruv>
PS C:\Users\guruv>
PS C:\Users\guruv>
PS C:\Users\guruv>
PS C:\Users\guruv> Test-Connection www.facebook.com
>>

Source      Destination     IPV4Address     IPV6Address                    Bytes    Time(ms)
------      -----------     -----------     -----------                    -----    --------
DESKTOPGV   www.facebook... 57.144.208.1                                    32       21
DESKTOPGV   www.facebook... 57.144.208.1                                    32       170
DESKTOPGV   www.facebook... 57.144.208.1                                    32       23
DESKTOPGV   www.facebook... 57.144.208.1                                    32       26


PS C:\Users\guruv> Test-Connection www.amazon.com
>>

Source      Destination     IPV4Address     IPV6Address                    Bytes    Time(ms)
```

```
------      -----------    -----------    -----------                    -----   --------
DESKTOPGV    www.amazon.com  108.159.19.102                                32      101
DESKTOPGV    www.amazon.com  108.159.19.102                                32      25
DESKTOPGV    www.amazon.com  108.159.19.102                                32      21
DESKTOPGV    www.amazon.com  108.159.19.102                                32      33


PS C:\Users\guruv> Test-Connection www.github.com
>>

Source      Destination    IPV4Address    IPV6Address                    Bytes   Time(ms)
------      -----------    -----------    -----------                    -----   --------
DESKTOPGV    www.github.com  20.207.73.82                                 32      40
DESKTOPGV    www.github.com  20.207.73.82                                 32      56
DESKTOPGV    www.github.com  20.207.73.82                                 32      93
DESKTOPGV    www.github.com  20.207.73.82                                 32      195


PS C:\Users\guruv> Test-Connection www.cisco.com
>>

Source      Destination    IPV4Address    IPV6Address                    Bytes   Time(ms)
------      -----------    -----------    -----------                    -----   --------
DESKTOPGV    www.cisco.com   23.60.108.111                                32      37
DESKTOPGV    www.cisco.com   23.60.108.111                                32      72
DESKTOPGV    www.cisco.com   23.60.108.111                                32      85
DESKTOPGV    www.cisco.com   23.60.108.111                                32      38


PS C:\Users\guruv> nslookup www.cisco.com
>>
```

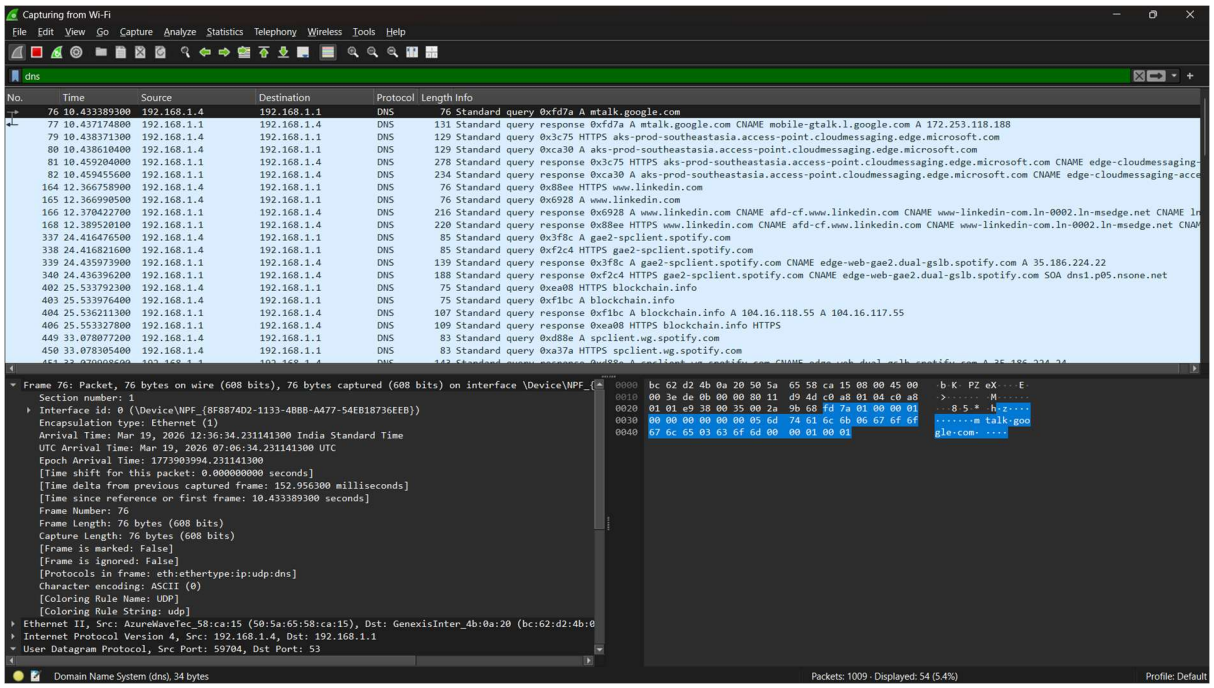
```
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    e2867.dsca.akamaiedge.net
Addresses:  2600:1417:53:193::b33
          2600:1417:53:1a9::b33
          23.60.108.111
Aliases:  www.cisco.com
          www.cisco.com.akadns.net
          wwwds.cisco.com.edgekey.net
          wwwds.cisco.com.edgekey.net.globalredir.akadns.net

PS C:\Users\guruv> nslookup www.github.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    github.com
Address:  20.207.73.82
Aliases:  www.github.com

PS C:\Users\guruv> nslookup www.amazon.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    e15316.dsca.akamaiedge.net
Addresses:  2600:1417:53:1ab::3bd4
```

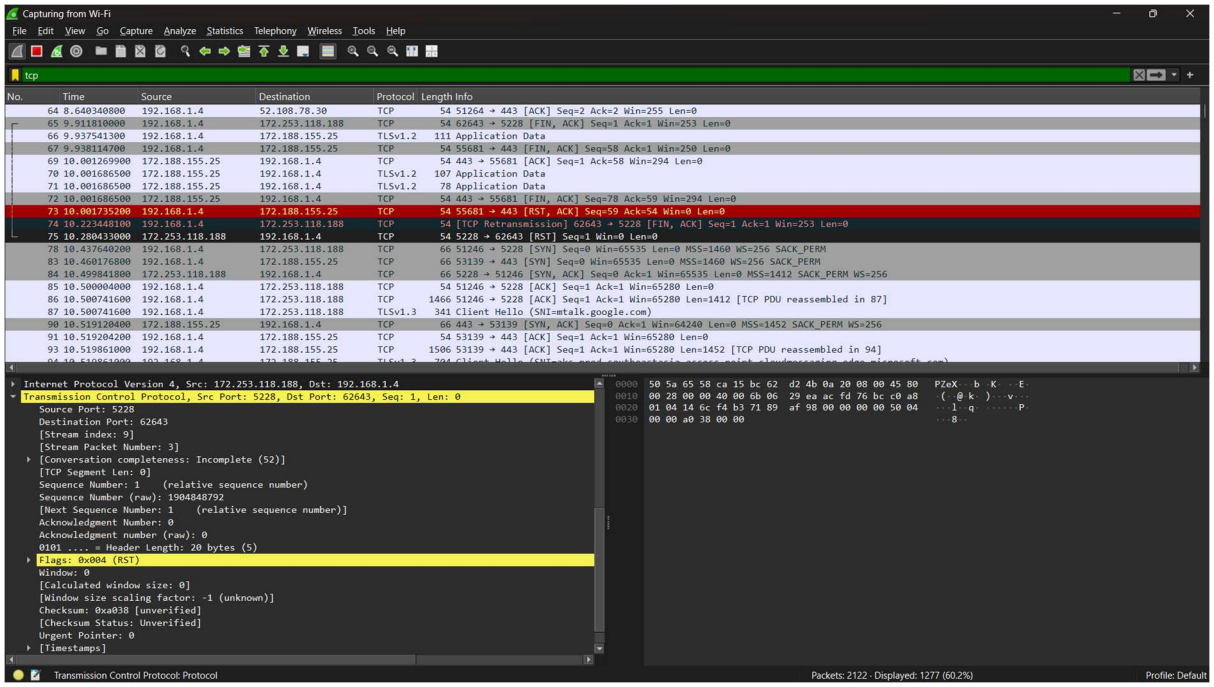
```
        2600:1417:53:1bb::3bd4
        23.60.110.84
Aliases:  www.amazon.com
        tp.47cf2c8c9-frontier.amazon.com
        www.amazon.com.edgekey.net

PS C:\Users\guruv> nslookup www.facebook.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    star-mini.c10r.facebook.com
Addresses:  2a03:2880:f368:1:face:b00c:0:25de
        57.144.208.1
Aliases:  www.facebook.com

PS C:\Users\guruv> nslookup www.google.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    www.google.com
Addresses:  2404:6800:4007:802::2004
        142.251.150.119
        142.251.151.119
        142.251.152.119
        142.251.153.119
        142.251.154.119
        142.251.155.119
```

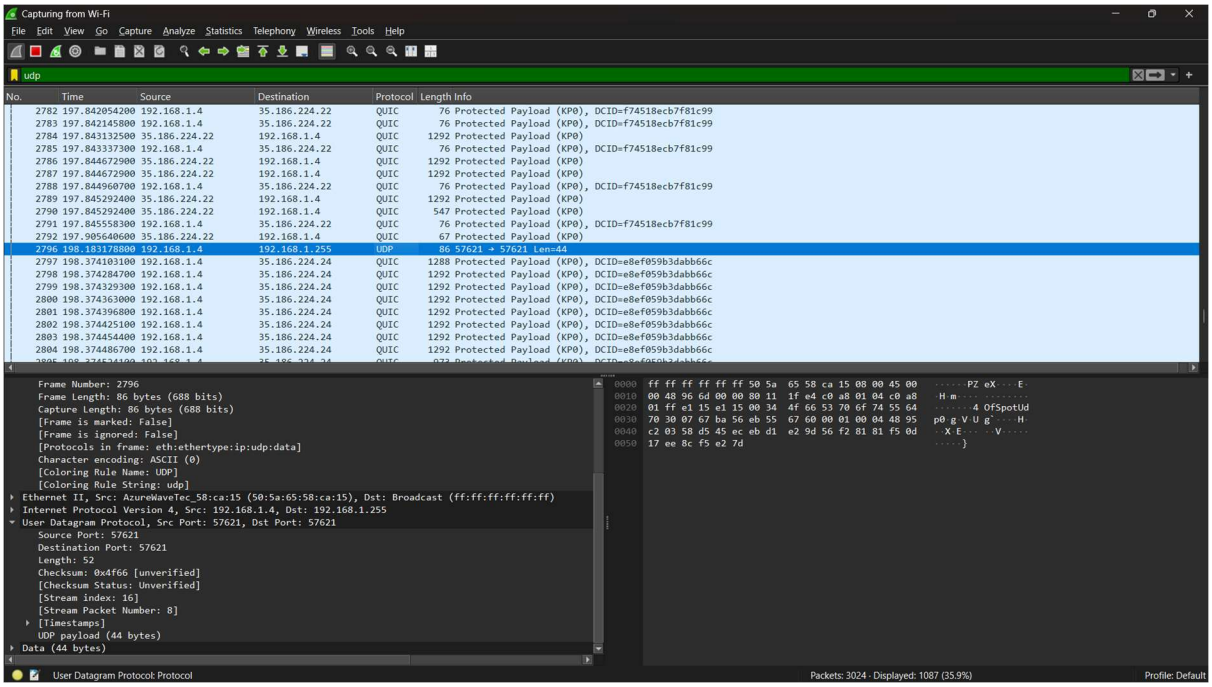
142.251.156.119

142.251.157.119

2)







3) vignesh@DESKTOPGV:~$ traceroute google.com

traceroute to google.com (142.251.220.110), 30 hops max, 60 byte packets

1 DESKTOPGV.mshome.net (172.23.96.1) 0.472 ms 0.433 ms 0.425 ms

2 192.168.1.1 (192.168.1.1) 5.797 ms 5.292 ms 5.979 ms

3 172.17.0.1 (172.17.0.1) 9.964 ms 9.947 ms 11.783 ms

4 * * *

5 172.16.178.1 (172.16.178.1) 30.287 ms 31.153 ms 31.817 ms

6 103.48.181.250 (103.48.181.250) 23.113 ms 21.978 ms 29.948 ms

7 * * *

8 * * *

9 142.250.235.104 (142.250.235.104) 24.582 ms 142.251.55.62 (142.251.55.62) 19.650 ms 142.251.55.224 (142.251.55.224) 26.060 ms

10 142.250.208.230 (142.250.208.230) 19.616 ms 142.251.55.223 (142.251.55.223) 20.151 ms 20.103 ms

11 hkg07s52-in-f14.1e100.net (142.251.220.110) 20.657 ms 172.253.70.167 (172.253.70.167) 29.432 ms pnmaaa-be-in-f14.1e100.net (142.251.220.110) 18.662 ms

vignesh@DESKTOPGV:~$ traceroute embedur.ai

traceroute to embedur.ai (162.159.137.54), 30 hops max, 60 byte packets

1 DESKTOPGV.mshome.net (172.23.96.1) 0.680 ms 0.585 ms 0.565 ms

2 192.168.1.1 (192.168.1.1) 3.105 ms 3.083 ms 2.028 ms

3 172.17.0.1 (172.17.0.1) 8.704 ms 15.109 ms 11.150 ms

4 160.30.102.74 (160.30.102.74) 38.252 ms 39.216 ms *

5 172.16.178.1 (172.16.178.1) 37.054 ms 36.175 ms 36.648 ms

6 103.48.181.250 (103.48.181.250) 35.298 ms 32.985 ms 33.170 ms

7 nsg-static-041.158.72.182.airtel.in (182.72.158.41) 35.977 ms 37.778 ms 23.895 ms

8 116.119.36.60 (116.119.36.60) 19.742 ms 25.609 ms 25.318 ms

9 * * *

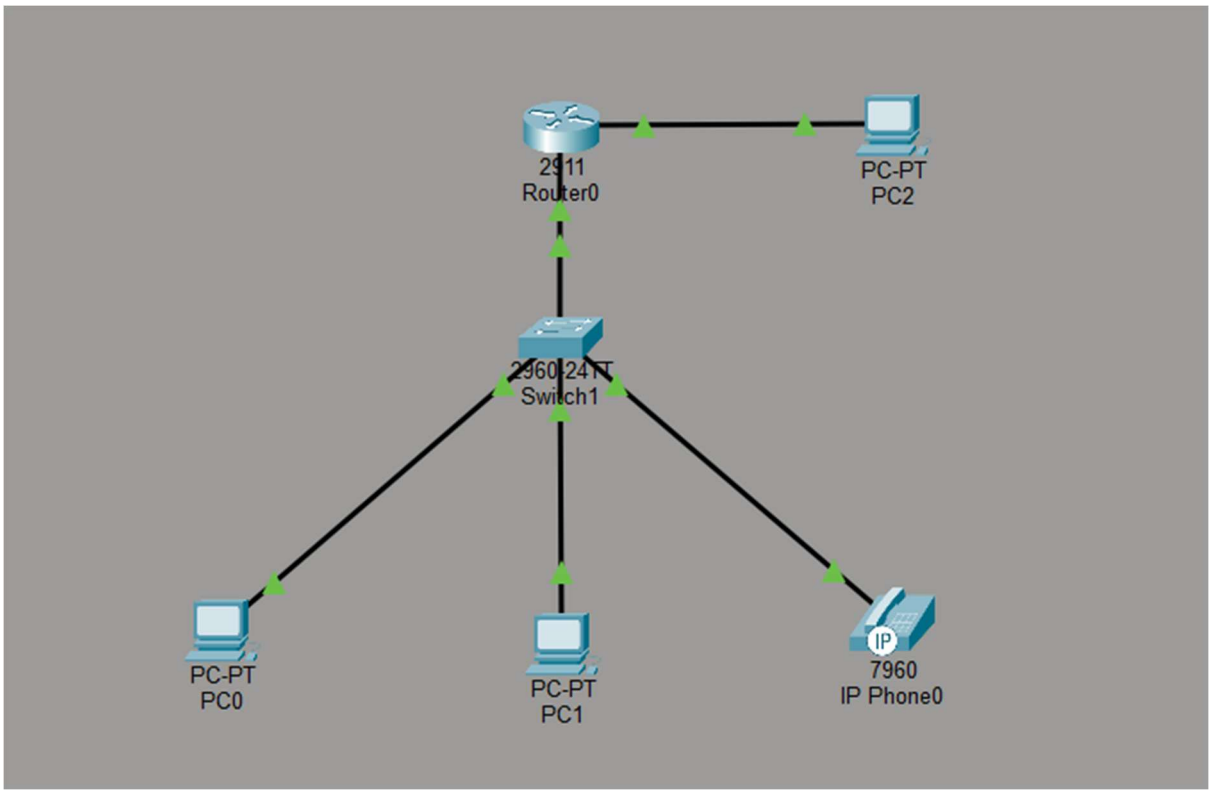
10 * 182.79.161.171 (182.79.161.171) 28.714 ms *

11 162.158.52.19 (162.158.52.19) 24.937 ms 162.158.52.35 (162.158.52.35) 72.686 ms 72.552 ms

12 162.159.137.54 (162.159.137.54) 22.948 ms 19.008 ms 19.181 ms

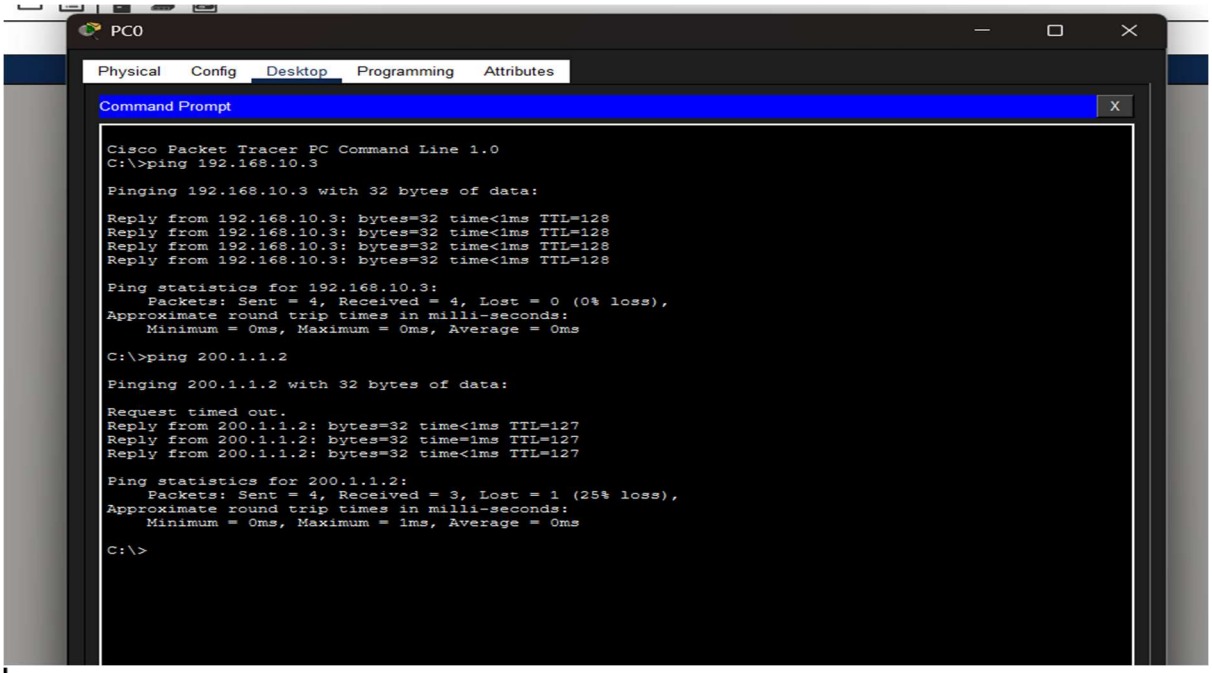
vignesh@DESKTOPGV:~$

CISCO PACKET TRACER ASSIGNMENTS (5,6,7,8,10)



```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name DATA
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name VOICE
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name MGMT
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport voice vlan 20
Switch(config-if)#interface fa0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#interface vlan 99
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

Switch(config-if)#ip address 192.168.99.2 255.255.255.
                                          ^
% Invalid input detected at '^' marker.

Switch(config-if)#ip address 192.168.99.2 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#ip default-gateway 192.168.99.1
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
```

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.99
Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

Router(config-subif)#interface g0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#
```




```
Switch>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   DATA                             active    Fa0/1, Fa0/2, Fa0/3
20   VOICE                            active    Fa0/3
99   MGMT                             active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>
```

```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/4       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/4       1-1005

Port        Vlans allowed and active in management domain
Fa0/4       1,10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/4       1,10,20,99

Switch>
```

9)

In a network where VLAN 10 and VLAN 20 are configured on a switch, devices in different VLANs cannot communicate with each other by default because VLANs operate at Layer 2 (Data Link Layer) and isolate broadcast domains. This means that traffic from one VLAN is not forwarded to another VLAN unless a Layer 3 device is introduced. The issue arises when there is no inter-VLAN routing configured, even though VLANs and ports are correctly assigned. To resolve this problem, a router or

Layer 3 switch must be used to perform Inter-VLAN Routing, typically through methods like Router-on-a-Stick, where subinterfaces are created for each VLAN and traffic is routed between them. Once routing is enabled and proper gateway addresses are configured on the end devices, communication between VLAN 10 and VLAN 20 becomes possible.

11)

An extended ACL was implemented to block HTTP traffic (TCP port 80) while allowing all other traffic.

```
access-list 101 deny tcp any any eq 80
access-list 101 permit ip any any

interface g0/0
ip access-group 101 in
```

This configuration ensures that HTTP traffic is denied at the router's inside interface.

12)

```
access-list 1 permit 192.168.10.0 0.0.0.255

interface g0/0
ip access-group 1 in
```

This type of ACL filters traffic based only on the source IP address.

13)

Extended ACL functionality was implemented to block specific application traffic. HTTP traffic was blocked as shown below:

```
access-list 102 deny tcp any any eq 80
access-list 102 permit ip any any

interface g0/0
ip access-group 102 in
```

Optionally, FTP traffic can also be blocked:

```
access-list 102 deny tcp any any eq 21
```

This demonstrates filtering based on port numbers using extended ACLs.

14)

NAT using PAT was implemented to allow multiple internal devices to share a single public IP address.

```
interface g0/0
ip nat inside

interface g0/1
ip nat outside
ip address 200.1.1.1 255.255.255.0
no shutdown
```

access-list 1 permit 192.168.0.0 0.0.255.255

ip nat inside source list 1 interface g0/1 overload

This configuration translates private IP addresses to a public IP, enabling communication with external networks.

15) **IPERF FROM WSL TO CMD (LOCALHOST)**

WSL OP

vignesh@DESKTOPGV:~$ sudo apt install iperf3 -y

vignesh@DESKTOPGV:~$ iperf3 -s

Server listening on 5201 (test #1)

Accepted connection from 172.23.96.1, port 58070

[5] local 172.23.98.200 port 5201 connected to 172.23.96.1 port 58071

[ID] Interval Transfer Bitrate

[5] 0.00-1.00 sec 743 MBytes 6.23 Gbits/sec

[5] 1.00-2.00 sec 773 MBytes 6.48 Gbits/sec

[5] 2.00-3.00 sec 772 MBytes 6.48 Gbits/sec

[5] 3.00-4.00 sec 747 MBytes 6.26 Gbits/sec

[5] 4.00-5.00 sec 698 MBytes 5.86 Gbits/sec

[5] 5.00-6.00 sec 754 MBytes 6.32 Gbits/sec

[5] 6.00-7.00 sec 744 MBytes 6.24 Gbits/sec

[5] 7.00-8.00 sec 735 MBytes 6.16 Gbits/sec

[5] 8.00-9.00 sec 748 MBytes 6.27 Gbits/sec

[5] 9.00-9.17 sec 127 MBytes 6.31 Gbits/sec

- -

[ID] Interval Transfer Bitrate

[5] 0.00-9.17 sec 6.68 GBytes 6.26 Gbits/sec receiver

Server listening on 5201 (test #2)

CMD OP

```
PS C:\Users\guruv\Downloads\iperf3.1.4_64> .\iperf3.exe -c 172.23.98.200
Connecting to host 172.23.98.200, port 5201
[  4] local 172.23.96.1 port 58071 connected to 172.23.98.200 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec   687 MBytes  5.76 Gbits/sec
[  4]   1.00-2.00   sec   706 MBytes  5.92 Gbits/sec
[  4]   2.00-3.00   sec   702 MBytes  5.89 Gbits/sec
[  4]   3.00-4.00   sec   709 MBytes  5.95 Gbits/sec
[  4]   4.00-5.00   sec   655 MBytes  5.49 Gbits/sec
[  4]   5.00-6.00   sec   652 MBytes  5.47 Gbits/sec
[  4]   6.00-7.00   sec   697 MBytes  5.85 Gbits/sec
[  4]   7.00-8.00   sec   668 MBytes  5.60 Gbits/sec
[  4]   8.00-9.00   sec   672 MBytes  5.64 Gbits/sec
[  4]   9.00-10.00  sec   693 MBytes  5.81 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  6.68 GBytes  5.74 Gbits/sec                  sender
[  4]   0.00-10.00  sec  6.68 GBytes  5.74 Gbits/sec                  receiver

iperf Done.
```